

- # Definition: - i) Differential Equation
 ii) Ordinary Differential Equation
 iii) Order and Degree of a DE.
 with example.

Formation of DE

1. Find the DE of all straight lines at unit distance from the origin.
2. Find the DE of the family of circles

$$x^2 + y^2 + 2gx + 2fy + c = 0$$
3. Find the DE of all circles which have their centres on X-axis and have a given radius.
4. Form the DE of the circle represented by $y^2 - 2ay + x^2 = a^2$, where a being arbitrary constant.
5. Find the DE of all circles passing through the origin and having their centres on the X-axis.
6. Find the DE of all circles of radius a .

7. Find the DE of all straight lines passing through the origin.
8. Find the DE of all straight lines not passing through the origin.
9. Form the DE of the family of parabolas with focus at the origin and axes being the x-axis.

Separation of Variables

$$1. \frac{dy}{dx} = \frac{1+y^2}{1+x^2}$$

$$2. \text{ special } a) \cos(x+y) dy = dx$$

$$b) \sin^{-1}\left(\frac{dy}{dx}\right) = x+y \Rightarrow \sin(x+y) = \frac{dy}{dx}$$

$$3. (x-4)y^4 dx - x^3(y^2-3) dy = 0$$

$$4. \text{ Solve the initial value problem } x \sin y dx + (x^2+1) \cos y dy = 0, y(1) = \frac{\pi}{2}$$

$$5. x(y^2+1) dx + y(x^2+1) dy = 0$$

Homogeneous DE

1. Define - Homogeneous DE.

2. $(x+y)dy + (x-y)dx = 0$

3. Solve the IVP $(x^r + y^r)dx - 2xy dy = 0, y(1) = 2$

4. $(x^3 + y^3)dx - 3xy^2 dy = 0$

5. $(2xy + 3y^2)dx + (2xy + x^2)dy = 0$

6. $\frac{dy}{dx} = \frac{3y^2 + x^2}{2xy}$

7. $(x^3 + y^3 \sqrt{x^2 + y^2})dx - xy \sqrt{x^2 + y^2} dy = 0$

8. Solve the initial value problem

9. $y' = \frac{y + \sqrt{x^2 + y^2}}{x}, y(1) = 0$

OR, $(x dy - y dx) = \sqrt{x^2 + y^2} dx$

OR, $x dy - (y + \sqrt{x^2 + y^2}) dx = 0$

9. $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$

10. Using the transformation $y = vx$ find the solution of $(x^r + y^r)dy = xy dx$

Exact DE

1. Define - Exact DE
2. Theorem:- State and prove the necessary and sufficient condition for the exactness of a differential equation

$$M(x,y) dx + N(x,y) dy = 0$$
3. Solve the initial value problem

$$(2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0, y(0) = 2$$
4. $(\sin x \cos y + e^{2x}) dx + (\cos x \sin y + \tan y) dy = 0$
5. $3x(xy-2) dx + (x^3+2y) dy = 0$
6. $(3x^2+4xy) dx + (2x^2+2y) dy = 0$
7. Show that xy is an integrating factor of $(3y+4xy^2) dx + (2x+3x^2y) dy = 0$

Non-exact DE

1. $(x^3 - 2y^2) dx + 2xy dy = 0$
2. $(2y^2 + 3x) dx + 2xy dy = 0$
3. $(x^2 + y^2 + x) dx + xy dy = 0$

4. $(5xy + 4y^r + 1) dx + (x^r + 2xy) dy = 0$

5. $(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$

Bernoulli DE

1. Define - Bernoulli DE.

2. $\frac{dy}{dx} + y = xy^3$

3. $\frac{dy}{dx} - y = xy^5$

4. $\frac{dy}{dx} + \frac{3y}{x} = 6y^r$

5. $\frac{dy}{dx} = x^3 y^3 - xy$

6. $\frac{dy}{dx} + \frac{y}{x} \log y = \frac{y}{x^2} (\log y)^2$

7. IVP: $\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}, y(1) = 2$

8. Write the DE $(x - y^r) dx + 2xy dy = 0$ into Bernoulli's DE form and then solve it.

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Linear DE

1. Define - Linear DE

$$2. \frac{dy}{dx} + \frac{3y}{x} = 6x^2$$

$$3. \frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$$

$$4. \text{Solve the IVP } xy' + y = e^x, y(1) = 2$$

$$5. \frac{dy}{dx} + 3y = 3x^2 e^{-3x}$$

$$6. \text{Solve the IVP } (x^2+1) \frac{dy}{dx} + 4xy = x, y(2) = 0$$

Higher Order Higher Degree (Homogeneous)

$$1. \frac{d^2y}{dx^2} - 6 \frac{dy}{dx} + 25y = 0$$

$$2. \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$$

Non-homogeneous

Operator Method

$$1. y'' - 4y = e^x$$

$$2. D^2y + 4Dy + y = \sin 2x$$

3. $D^r y + y = \cos 2x$

4. $(D^r + 4)y = \cos^r x$

5. $(D^3 - D^r - 6D)y = 1 + x^r$

6. $(D^r + 4)y = x^r e^{2x}$

7. $(D^r - 9)y = 2x e^{3x}$

8. $y''' - 2y' + 4y = e^x \cos x$

Undetermined Co-efficients

1. $D^r y - 2Dy - 3y = 2e^x - 10 \sin x$

2. $y'' + 4y' + 9y = x^r + 3x$

Variation of Parameters

1. $y'' + y = \sec x$

2. $y'' + y = \tan x$

Solve $(D^r+4)y = x^r e^{2x}$

Solution:- Given that

$$(D^r+4)y = x^r e^{2x}$$

Let us consider

$$(D^r+4)y = 0 \quad \dots (1)$$

Then the auxiliary equation is

$$m^r+4=0$$

$$\Rightarrow m^r = -4$$

$$\Rightarrow m^r = 4i^r$$

$$\Rightarrow m = \pm 2i$$

The complementary function is

$$y_c = c_1 \cos 2x + c_2 \sin 2x$$

The particular integral is

$$y_p = \frac{1}{D^r+4} e^{2x} x^r$$

$$= \frac{e^{2x}}{(D+2)^r+4} x^r$$

$$= \frac{e^{2x}}{D^r+4D+4+4} x^r$$

$$\begin{aligned}
&= \frac{e^{2x}}{D^2 + 4D + 8} x^2 \\
&= \frac{e^{2x}}{8 \left(1 + \frac{4D}{8} + \frac{D^2}{8} \right)} x^2 \\
&= \frac{e^{2x}}{8 \left(1 + \frac{D}{2} + \frac{D^2}{8} \right)} x^2 \\
&= \frac{e^{2x}}{8} \left\{ 1 + \left(\frac{D}{2} + \frac{D^2}{8} \right) \right\}^{-1} x^2 \\
&= \frac{e^{2x}}{8} \left\{ 1 - \left(\frac{D}{2} + \frac{D^2}{8} \right) + \left(\frac{D}{2} + \frac{D^2}{8} \right)^2 - \dots \right\} x^2 \\
&= \frac{e^{2x}}{8} \left(1 - \frac{D}{2} - \frac{D^2}{8} + \frac{D^2}{4} \right) x^2 \\
&= \frac{e^{2x}}{8} \left(x^2 - \frac{1}{2} \cdot 2x - \frac{1}{8} \cdot \frac{2}{2} + \frac{1}{4} \cdot 2 \right) \\
&= \frac{e^{2x}}{8} \left(x^2 - x - \frac{1}{4} + \frac{1}{2} \right) \\
&= \frac{e^{2x}}{8} \left(x^2 - x + \frac{1}{4} \right)
\end{aligned}$$

Therefore, the general solution is

$$\begin{aligned}
y &= y_c + y_p \\
&= c_1 \cos 2x + c_2 \sin 2x + \frac{e^{2x}}{8} \left(x^2 - x + \frac{1}{4} \right)
\end{aligned}$$

where c_1 and c_2 are arbitrary constants.

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